

Upset Losses, Compliant Citizens: Evidence from Chinese Football

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Abstract

Do incidental emotional shocks affect political attitudes toward state authority? We link Chinese Super League football results to individual interview dates in the Chinese General Social Survey (2015, 2018, 2021, 2023) and exploit within-province-year variation in exposure to unexpected match outcomes. Upset losses by a respondent's provincial team reduce support for government non-interference with public criticism by 0.27 points on a five-point scale (0.24 standard deviations) in the one-day window. The effect attenuates over the following week and is absent for expected losses and unexpected wins. It is concentrated among respondents with high internet use and urban residence, and does not extend to generalized social trust or happiness. These findings indicate that transient negative emotions can increase acceptance of government control over political expression, extending the behavioral-spillover literature to political attitudes in an authoritarian context.

Keywords: incidental emotions, political attitudes, football, China, unexpected events

JEL Classification: D91, P26, Z20

1 Introduction

Citizens’ willingness to accept government control over political expression is a central feature of authoritarian governance, yet we know little about what shapes it beyond government actions themselves. Anti-corruption campaigns affect both economic outcomes such as poverty reduction (Han et al., 2022) and firm productivity (Kong et al., 2020), as well as political attitudes (Fang et al., 2025). We ask whether an emotional shock with no connection to politics can alter how citizens view state authority. Specifically, we study whether unexpected football losses affect public acceptance of government intervention in political expression in China.

Emotions triggered by events outside a decision domain can reshape behavior within it. Card and Dahl (2011) show that when a favored NFL team loses unexpectedly, domestic violence rises by roughly 10%, while expected losses have no effect; Cardazzi et al. (2024) extend this finding to basketball. The gap between realized and expected outcomes drives these effects (Kőszegi and Rabin, 2006). The same logic applies in financial markets (Edmans et al., 2007), judicial sentencing (Eren and Mocan, 2018), consumer spending (Cai and Li, 2026), tipping (Ge, 2018), and elections (Healy et al., 2010). A consistent pattern runs through these findings: the behavioral response matches the valence of the shock, and negative surprises matter more than positive ones. Yet all of this evidence comes from democracies, and in every case the spillover runs in the same direction as the shock. Whether negative emotions can *increase* acceptance of authority—reversing this same-valence pattern—remains untested.

We link the timing of Chinese Super League (CSL) matches to individual interview dates in the Chinese General Social Survey (CGSS). Following Card and Dahl (2011) and Ivandić et al. (2024), we classify match outcomes relative to pre-match betting odds. Identification comes from within-province-year variation in interview timing relative to match dates: respondents in the same province and year who are interviewed before versus after an upset loss serve as quasi-experimental comparisons, an application

of the Unexpected Event during Survey Design (UESD) framework (Muñoz et al., 2020).

Upset losses reduce support for government non-interference with public criticism by 0.272 points on a five-point scale ($p < 0.01$) in the one-day window, equivalent to 0.24 standard deviations. The effect fades to -0.165 ($p < 0.05$) at three days and -0.132 ($p < 0.10$) at seven days. Close losses and upset wins produce no comparable effects. The sharp initial response followed by rapid decay points to a transient emotional channel rather than a lasting attitude change.

Several findings help pin down the mechanism. The effect is concentrated among respondents with high internet use and urban *hukou* status, the groups most likely to follow match outcomes. This gradient is hard to square with stories in which upset losses proxy for broader provincial shocks. Upset losses do not affect generalized social trust or self-reported happiness, ruling out a simple negative-mood account (Schwarz and Clore, 1983). Instead, the specificity of the shift to attitudes toward government authority supports the appraisal-tendency framework, in which discrete emotions produce domain-specific rather than diffuse effects (Lerner and Keltner, 2001; Meier, 2022). The results survive alternative fixed-effects structures, controls for upset-loss severity, and randomization inference.

We contribute to three literatures. First, we extend the sports-emotions spillover literature to political attitudes. Existing studies document spillovers into violence, financial markets, sentencing, spending, and voting, but all in democratic settings and all following a same-valence pattern. The closest study to ours, Kikuta and Uesugi (2023), shows that close losses by African players' European clubs temporarily *decrease* trust in domestic leaders and increase protests, a same-valence response in a weak-state setting. We show that in an authoritarian context, the same type of shock can *increase* acceptance of government intervention, reversing the usual mapping. This complements evidence that negative emotions raise demand for government action (Algan et al., 2025), reduce support for political change (Meier et al., 2019), and alter compliance with state authority (Fochmann et al., 2025).

Second, we contribute to the study of political attitudes in China, where existing work emphasizes how government actions shape trust and compliance (Fang et al., 2025; Li and Zheng, 2025). We show that attitudes toward state control over expression respond to emotional cues orthogonal to government performance, suggesting that the demand side of authoritarian governance is more malleable than standard accounts of instrumental evaluation or cultural values would imply.

Third, we provide a proof of concept for using the UESD approach to study sensitive political attitudes in authoritarian settings, where survey experiments risk censorship or preference falsification, and where sporting outcomes offer a credible source of emotional variation.

The remainder of the paper proceeds as follows. Section 2 discusses background and motivation. Section 3 describes the research design. Section 4 presents results and heterogeneity analysis. Section 5 reports robustness checks. Section 6 concludes.

2 Background and Motivation

Our approach builds on the insight that sporting outcomes generate plausibly exogenous emotional shocks whose effects can be traced into other domains. Card and Dahl (2011) show that upset NFL losses, defeats when the team was favored, increase domestic violence by roughly 10%, while expected losses have no effect. The emotional response is driven by the gap between outcome and expectation (Kőszegi and Rabin, 2006), not by losing itself. This asymmetry between upset and expected outcomes appears across financial markets (Edmans et al., 2007), judicial sentencing (Eren and Mocan, 2018), consumer spending (Cai and Li, 2026), and tipping (Ge, 2018). In the political domain, Healy et al. (2010) find that college football victories boost local incumbent vote shares, Depetris-Chauvin et al. (2020) show that national team victories reshape ethnic identities in Africa, and Kikuta and Uesugi (2023) find that close losses by African players’ European clubs temporarily decrease trust in domestic leaders and trigger

protests. Closest to our setting, [Guo et al. \(2025\)](#) exploit the timing of the 2016 Rio Olympics and CGSS interviews to show that female gold medalists shift gender preferences in their home provinces, using the same survey data we employ.

Two features of this literature motivate our paper. First, every documented spillover follows a same-valence pattern: negative shocks produce negative downstream outcomes and positive shocks produce favorable ones. No study has tested whether negative emotions can *increase* acceptance of authority, a shift that would reverse this mapping. Second, the evidence comes almost entirely from democracies, and even the one cross-continental exception ([Kikuta and Uesugi, 2023](#)) finds a same-valence response in weakly institutionalized states. Recent work suggests the valence mapping may not be universal. [Algan et al. \(2025\)](#) find that experimentally induced negative emotions increase support for protectionist and redistributive policies in the United States. [Fochmann et al. \(2025\)](#) show that incidental positive emotions reduce tax compliance in Germany. Most directly, [Meier et al. \(2019\)](#) show that rain-induced negative emotions reduce support for political change in Swiss referenda, linking negative affect to preference for the status quo. A psychological account for these patterns comes from the compensatory control literature: when personal control is threatened, people turn to external systems of order, including government, to restore a sense of predictability ([Kay et al., 2008, 2009](#)). None of this work, however, examines an authoritarian setting, where the relevant margin is not demand for redistribution or the status quo on a ballot, but acceptance of government control over political expression.

3 Research Design

3.1 Institutional Setting: The Chinese Super League

The Chinese Super League (CSL) is the top tier of professional football in China. Founded in 2004, the league comprises 16 clubs, each based in a provincial capital or major city and closely identified

with its home province. In 2015, the State Council issued the National Plan for Chinese Football Reform and Development, designating football as a national strategic priority (Cockayne et al., 2022). Average match-day attendance reached approximately 18,000 during the early years of our sample and rose to over 24,000 by 2018, ranking the CSL among the best-attended football leagues in the world (Watanabe and Soebbing, 2017). All matches are broadcast on CCTV5, the only free national sports channel with nationwide coverage, as well as on major streaming platforms, and the league’s media rights for 2016–2026 sold for 165 million USD (Li et al., 2025). This wide exposure ensures that match results reach audiences well beyond stadium attendees. The combination of strong provincial identification with local clubs and broad media reach makes CSL outcomes a plausible source of short-lived emotional shocks for the general provincial population.

3.2 Data

Football data Following Card and Dahl (2011) and Ivandić et al. (2024), we use betting markets to measure pre-game expectations and classify match outcomes into emotional-cue categories. Pre-game odds define the benchmark against which fans evaluate realized outcomes (Kőszegi and Rabin, 2006), so a defeat when the team was favored generates a larger negative response than one that was expected (Card and Dahl, 2011; Edmans et al., 2007). We collect match-level data for all Chinese Super League (CSL) games from 2014 to 2023 from www.football-data.co.uk. The data provide the match date, home and away teams, full-time result (home win, draw, or away win), and pre-game decimal odds for each of the three outcomes.

We convert odds into implied probabilities and remove the bookmaker margin by normalizing inverse odds so that $p_H + p_D + p_A = 1$. Using the team’s implied win probability p_{win} , we classify each match from each team’s perspective as a *predicted win* ($p_{\text{win}} \geq 0.55$), *predicted loss* ($p_{\text{win}} < 0.45$), or *predicted close* (otherwise). These thresholds follow Ivandić et al. (2024), who apply the same cutoffs to European

football leagues.¹

Combining these predicted-outcome bins with realized results yields six mutually exclusive categories: *upset loss* (expected win, actual loss), *close loss* (close match, actual loss), *predicted loss* (expected loss, actual loss), and the corresponding win-side categories *upset win*, *close win*, and *predicted win*. Draws form a separate category and do not enter the win/loss classifications. An upset loss thus captures the negative surprise of a defeat when the team was favored, while a predicted loss carries little new information and should produce a weaker response (Kőszegi and Rabin, 2006; Card and Dahl, 2011).

Figure 1 plots the realized goal difference (home goals minus away goals) against the implied home-win probability for all CSL matches in our sample. The positive relationship validates betting odds as informative measures of pre-game expectations.

Household survey data We draw individual-level data from the Chinese General Social Survey (CGSS), a nationally representative repeated cross-section administered by the National Survey Research Center at Renmin University of China. The CGSS uses a multi-stage stratified probability sampling design covering all provinces in mainland China, and records the exact date of each face-to-face interview, which is essential for linking respondents to recent match outcomes in their province. We pool the 2015, 2018, 2021, and 2023 waves, which share a common set of attitudinal and demographic items.²

Our primary outcome is *support for government non-interference with public criticism*, measured by agreement with the statement “if someone publicly makes remarks criticizing the government, the

¹Card and Dahl (2011) classify NFL games using a point-spread threshold of 4 points, corresponding roughly to the 75th percentile of the absolute spread distribution in their data. Because football (soccer) betting markets quote three-way odds rather than point spreads, we adopt the probability-based classification of Ivandić et al. (2024), which maps onto an analogous distinction between games where one side is a clear favorite, games that are roughly evenly matched, and games where the team is a clear underdog.

²We exclude the 2017 wave, which does not record interview dates and therefore cannot be linked to match outcomes. The 2021 wave measures social trust on a four-point rather than five-point scale; accordingly, social trust regressions use only the 2015, 2018, and 2023 waves.

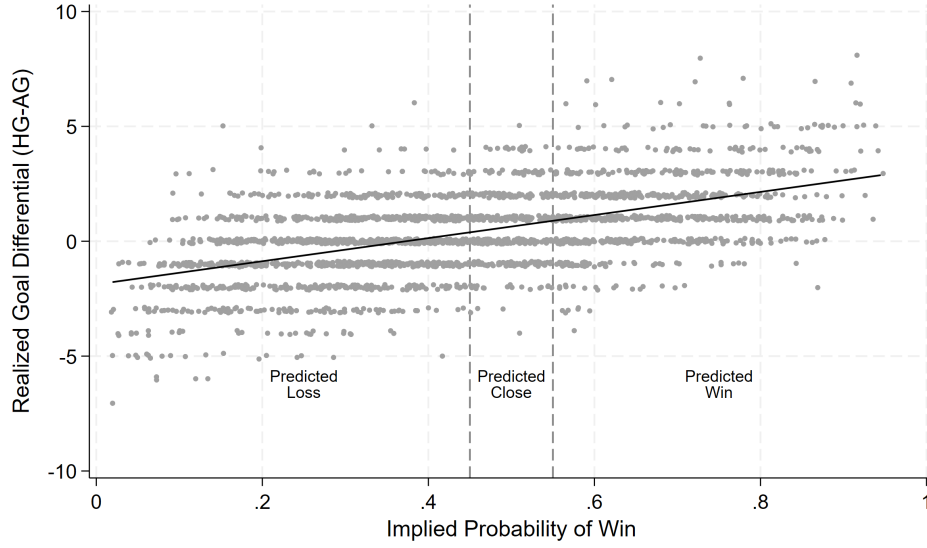


Figure 1: Betting Market Efficiency

Notes: Each point represents a Chinese Super League match, 2014 to 2023. The horizontal axis shows the implied home-win probability derived from normalized betting odds; the vertical axis shows the realized goal difference (home goals minus away goals). The solid line is a linear fit. Vertical dashed lines mark the classification thresholds at $p = 0.45$ and $p = 0.55$ that separate predicted losses, close matches, and predicted wins.

government should not interfere” (1 to 5 scale, where higher values indicate greater support for non-interference and lower values indicate greater acceptance of government intervention).

Individual-level controls include gender, Han ethnicity, years of education, marital status, Chinese Communist Party membership, rural *hukou* status, self-rated health, and the log of personal income. These variables capture the key demographic and socioeconomic characteristics that prior work has shown to predict political attitudes and subjective wellbeing in China (Fang et al., 2025; Booth et al., 2022; Zhang et al., 2025; Asadullah et al., 2018; Zhang and Churchill, 2020). We additionally control for television and internet use frequency (each on a 1 to 5 scale) to account for differential media exposure that might correlate with both awareness of match outcomes and attitudinal responses. All regressions use CGSS sampling weights.

Province-level variables We merge province-level variables from the National Bureau of Statistics of China at the province-year level, including the log of GDP per capita, log population, and the tertiary-sector share (Han et al., 2022). These variables are reported descriptively in the summary statistics. They are not included separately in the main regressions because province-by-year fixed effects absorb all province-year variation.

3.3 Descriptive Statistics

Table 1 reports descriptive statistics for the estimation sample. On average, respondents score 2.58 on the five-point scale of support for government non-interference with public criticism, indicating that the typical response lies between “somewhat disagree” and “neither agree nor disagree.” The football shock variables in Panel B show that upset losses are rare events in a one-day window (mean = 0.003), which motivates our use of short pre-interview windows spanning up to one week in the main analysis. Individual characteristics in Panel C are broadly consistent with the composition of the adult survey sample: 50% female, 96% Han, and roughly 10 years of schooling on average.

Figure 2 illustrates the geographic distribution of our key variables. Panel (a) shows the average annual number of upset losses by province, concentrated in provinces with active CSL teams, primarily in eastern and central China. Panel (b) shows the province-level mean of support for government non-interference with public criticism. The maps highlight two features of the empirical setting. First, there is substantial cross-province variation in the attitudinal outcome, which is absorbed by our province fixed effects. Second, the provinces most exposed to football shocks (e.g., Shandong, Guangdong, and Hebei) span a range of attitudinal levels, mitigating concerns that the results are driven by a single type of province.

Table 1: Summary Statistics

Variable	N	Mean	Std. Dev.	Min	Max
<i>Panel A: Outcome Variables</i>					
Support for non-interference with public criticism (1–5)	20,892	2.581	1.115	1	5
Social trust (1–5)	16,663	3.504	0.976	1	5
Life happiness (1–5)	19,463	3.921	0.796	1	5
<i>Panel B: Football Shock Variables (1-day window)</i>					
Upset loss	20,892	0.003	0.051	0	1
Close loss	20,892	0.001	0.038	0	1
Upset win	20,892	0.020	0.142	0	1
Predicted win	20,892	0.027	0.166	0	2
Close match	20,892	0.015	0.120	0	1
Predicted loss	20,892	0.099	0.308	0	2
<i>Panel C: Individual Controls</i>					
Female (=1)	20,892	0.501	0.500	0	1
Han ethnicity (=1)	20,892	0.964	0.187	0	1
Education (years)	20,892	9.796	4.621	0	19
Married (=1)	20,892	0.746	0.435	0	1
CCP member (=1)	20,892	0.122	0.327	0	1
Rural hukou (=1)	20,892	0.591	0.492	0	1
Self-rated health (1–5)	20,892	3.654	1.053	1	5
Log personal income	20,892	8.719	3.822	0	16
<i>Panel D: Media Controls</i>					
TV use frequency (1–5)	20,892	3.482	1.188	1	5
Internet use frequency (1–5)	20,892	3.288	1.679	1	5
<i>Panel E: Province-level Variables (descriptive only)</i>					
Log GDP per capita	20,481	11.202	0.434	10	12
Log population	20,481	8.609	0.611	7	9
Tertiary sector share	20,481	54.894	10.724	39	85

Notes: Sample restricted to CGSS respondents from the 2015, 2018, 2021, and 2023 waves with non-missing values for all individual controls, media controls, and sampling weights, matching the estimation sample used in the main regressions. Social trust is available only for the 2015, 2018, and 2023 waves (the 2021 wave uses a four-point scale). Football shock variables are counts within a 1-day pre-interview window $[d-1, d-1]$, excluding the interview day. All statistics are weighted using CGSS sampling weights.

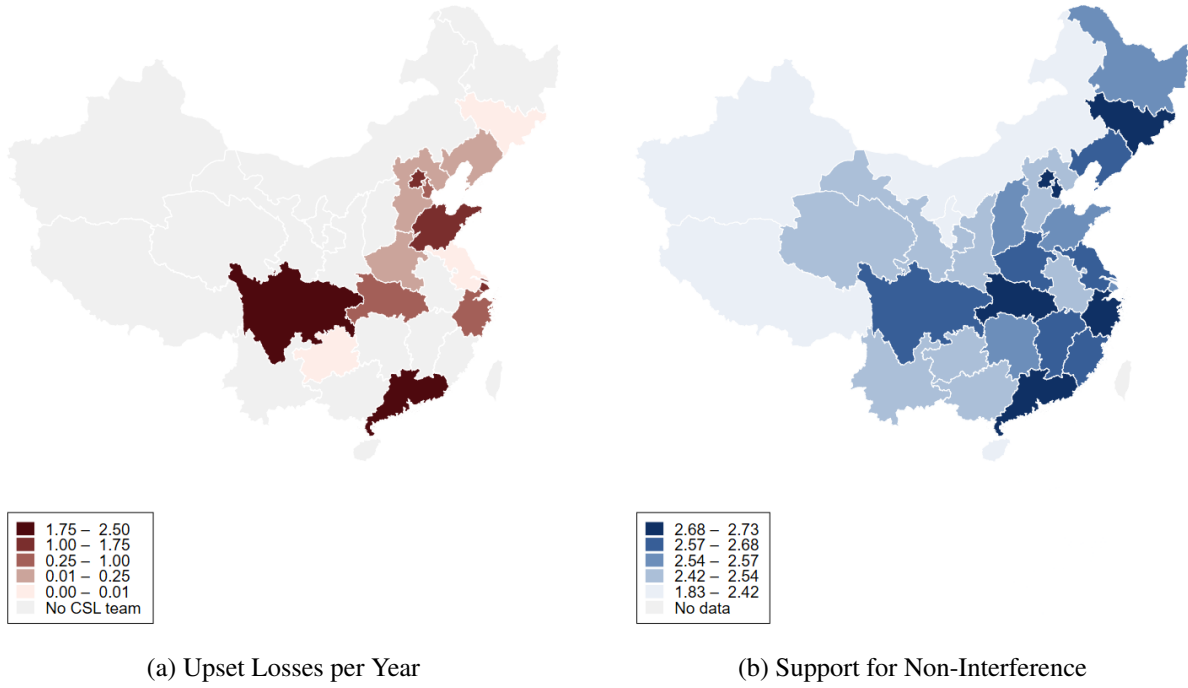


Figure 2: Geographic Variation in Upset Losses and Support for Non-Interference

Notes: Province-level maps of China. Figure 2a shows the average annual number of upset losses (unexpected defeats based on pre-match betting odds) by province, 2015 to 2023. Figure 2b displays the province-level weighted mean of support for government non-interference with public criticism (1 to 5 scale) from the CGSS. Grey provinces indicate no CSL team (panel a) or no CGSS data (panel b).

3.4 Empirical Strategy

We exploit quasi-random variation in the timing and outcomes of CSL matches relative to CGSS interview dates within the same province and year. Provinces with CSL teams differ systematically from those without, but the exact date on which a team plays, and whether it wins or loses relative to expectations, is plausibly orthogonal to the date on which a given respondent is interviewed, conditional on province-by-year fixed effects. This is a specific application of the Unexpected Event during Survey Design (UESD) framework (Muñoz et al., 2020). We estimate:

$$Y_{ipt} = \beta_1 \text{UpsetLoss}_{p,W} + \beta_2 \text{CloseLoss}_{p,W} + \beta_3 \text{UpsetWin}_{p,W} + \gamma' \mathbf{C}_{p,W} + \delta' \mathbf{X}_{ipt} + \alpha_{pt} + \varepsilon_{ipt}, \quad (1)$$

where Y_{ipt} is support for government non-interference with public criticism for individual i in province p interviewed in year t . The variables $\text{UpsetLoss}_{p,W}$, $\text{CloseLoss}_{p,W}$, and $\text{UpsetWin}_{p,W}$ count the number of upset losses, close losses, and upset wins experienced by the province’s CSL team(s) in the W -day window preceding the interview, specifically over the interval $[d-W, d-1]$, where d is the interview date. We exclude the interview day itself to ensure that only pre-interview shocks enter the specification. The omitted category comprises respondents whose provincial team experienced only predicted outcomes (predicted wins, close wins, or predicted losses) or no matches during the window. Following [Card and Dahl \(2011\)](#), we include predicted-outcome controls $\mathbf{C}_{p,W}$, consisting of counts of predicted wins, all close matches, and predicted losses within the same window. These controls ensure that the shock coefficients β_1 to β_3 capture the effect of deviations from expected outcomes rather than the match schedule itself. With total close matches held constant, β_2 captures the effect of a close loss relative to a close win.

The vector $\mathbf{X}_{ipt}$ includes individual-level controls: gender, Han ethnicity, years of education, marital status, CCP membership, rural *hukou* status, self-rated health, log personal income, and television and internet use frequency. The term α_{pt} denotes province-by-year fixed effects, which absorb all time-invariant province characteristics as well as province-specific annual factors that may correlate with team performance and respondents’ attitudes. Identification therefore relies entirely on within-province-year variation: we compare respondents in the same province and year who happen to be interviewed on dates that differ in their exposure to recent unexpected match outcomes. Standard errors are clustered at the province level. All regressions are weighted using CGSS sampling weights.

The coefficient of primary interest is β_1 . If an upset loss elicits negative arousal that increases demand for order and external control, we expect $\beta_1 < 0$: an upset loss should reduce support for the proposition that the government should not interfere with public criticism, i.e., increase acceptance of

government intervention. The coefficient β_2 on close losses serves as a benchmark: these are losses from roughly evenly matched games that carry little surprise and should therefore produce a weaker emotional response. The coefficient β_3 on upset wins tests whether positive surprises generate comparable short-run spillovers.

We estimate equation (1) across three window widths, $W \in \{1, 3, 7\}$ days. Shorter windows ($W = 1, 3$) isolate acute responses to recent match outcomes, while the 7-day window allows us to examine whether these effects persist within the following week. The comparison across windows is informative about the nature of the effect: a sharp response concentrated in the shortest windows is consistent with a transient emotional cue rather than a durable change in attitudes ([Card and Dahl, 2011](#)).

Identifying assumption. The validity of our approach requires that, conditional on province-by-year fixed effects, the timing and outcomes of CSL matches are uncorrelated with unobserved determinants of individual attitudes. Two features of our setting support this assumption. First, the CSL schedule is set by the Chinese Football Association before the season begins and is not influenced by provincial socioeconomic conditions or survey logistics. Second, CGSS interview dates are determined by fieldwork protocols, including interviewer availability, respondent scheduling, and sampling sequences, which are plausibly independent of match outcomes. We probe this assumption in several ways. We estimate a lead specification that replaces the pre-interview window $[d - W, d - 1]$ with the post-interview window $[d + 1, d + W]$; significant effects of future match outcomes on current attitudes would indicate confounding pre-trends or spurious seasonality.

4 Results

4.1 Main Results

Table 2 reports estimates of equation (1) across three pre-interview windows. In the one-day window (column 1), an upset loss reduces support for non-interference by 0.272 points on the five-point scale ($p < 0.01$), equivalent to 0.24 standard deviations. The effect attenuates but remains significant at three days (-0.165 , $p < 0.05$) and seven days (-0.132 , $p < 0.10$). This decay pattern indicates a short-run emotional channel ([Card and Dahl, 2011](#); [Lerner et al., 2015](#)). Because the estimate is an intent-to-treat effect at the province level, averaging over many residents who do not follow the local team, the effect on attentive fans is likely substantially larger.

Close losses show no comparable negative effect in any window. This contrast indicates that deviations from expectations, rather than losses per se, drive the result, consistent with reference-dependent utility ([Kőszegi and Rabin, 2006](#)). Upset wins are likewise insignificant, indicating that the spillover is asymmetric: negative surprises exert a stronger immediate effect than positive ones, in line with the well-documented negativity bias in affective responses ([Baumeister et al., 2001](#)).

Table 2: Effect of Unexpected Match Outcomes on Support for Government Non-Interference

	(1) Day 1	(2) Day 3	(3) Day 7
Upset loss	-0.272*** (0.078)	-0.165** (0.067)	-0.132* (0.068)
Close loss	0.027 (0.093)	0.107 (0.142)	0.107 (0.062)
Upset win	-0.043 (0.067)	-0.048 (0.043)	-0.016 (0.038)
Mean of dep. var.	2.581	2.581	2.581
Province $\times$ year FE	Yes	Yes	Yes
Individual controls	Yes	Yes	Yes
Media controls	Yes	Yes	Yes
N	20,892	20,892	20,892
R ²	0.053	0.054	0.054

Notes: The dependent variable is support for government non-interference with public criticism (1–5, with higher values indicating greater support for non-interference). Each column uses a pre-interview window $[d - W, d - 1]$, where $W \in \{1, 3, 7\}$ and d is the interview date. All columns include province $\times$ year fixed effects, individual controls, media controls, and Card-style predicted-category controls. Standard errors are clustered at the province level. Regressions are weighted using CGSS sampling weights. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

4.2 Heterogeneity Analysis

Table 3 examines whether the effect varies with proxies for exposure to match outcomes. Panel A splits the sample by internet use. Among high-internet respondents, upset losses reduce support for non-interference across all three windows (-0.257 , -0.206 , and -0.152), with coefficients that decay monotonically and remain statistically significant. Among low-internet respondents, the estimates are smaller and insignificant. This gradient is consistent with an information-exposure channel: the attitudinal effect is concentrated among individuals more likely to observe and discuss match outcomes. This finding aligns with evidence that social media use shapes subjective wellbeing through social comparison (Bao et al., 2021).

Panel B splits the sample by *hukou* status. The estimates are larger among urban respondents (-0.542 in the 1-day window, -0.134 at 7 days), while the pattern for rural respondents is weaker and less stable. Together, Panels A and B are difficult to reconcile with stories in which upset losses proxy for broader

provincial shocks, since such shocks should not systematically vary with internet use or urban residence. The results instead indicate that the emotional cue operates among individuals more likely to learn about the match outcome.

Panel C shows that the effect is present for both men and women, though with different temporal profiles: male respondents show significant effects in the 1-day and 3-day windows (-0.249 and -0.229), while female respondents respond in the 1-day and 7-day windows (-0.299 and -0.213). Women’s responsiveness is notable given that football fandom in China is disproportionately male, suggesting that the effect extends beyond direct viewership to household interactions or broader media exposure. Across all panels, close losses and upset wins show no systematic pattern, reinforcing the interpretation that the surprise component of negative outcomes is central.

4.3 Event-Study Evidence

Figure 3 replaces the cumulative windows with binned indicators for upset losses occurring 1 to 3 and 4 to 7 days before the interview, alongside placebo lead bins (1 to 3 and 4 to 7 days after) that should be zero under the identifying assumption. The 1-to-3-day lag bin shows a significant negative effect (-0.135 , $p < 0.001$), while the 4-to-7-day lag remains negative but is no longer statistically significant. Both placebo leads are indistinguishable from zero, providing reassurance against pre-trends or reverse causality.

5 Robustness Checks

This section examines sensitivity to alternative outcomes, fixed-effects structures, measures of shock intensity, and inference procedures.

Table 3: Heterogeneity in the Effect of Football Shocks on Support for Non-Interference

<i>Panel A: High vs. Low Internet Use</i>						
	High Internet Use			Low Internet Use		
	Day 1	Day 3	Day 7	Day 1	Day 3	Day 7
Upset loss	-0.257*** (0.044)	-0.206*** (0.062)	-0.152** (0.070)	-0.294 (0.231)	0.018 (0.075)	-0.041 (0.045)
Close loss	-0.438*** (0.163)	0.167 (0.212)	0.141 (0.159)	0.246*** (0.052)	0.070 (0.101)	0.073 (0.100)
Upset win	-0.064 (0.078)	-0.042 (0.059)	-0.005 (0.051)	0.013 (0.083)	-0.064 (0.047)	-0.024 (0.047)
N	10,734	10,734	10,734	10,158	10,158	10,158
<i>Panel B: Urban vs. Rural Hukou</i>						
	Urban			Rural		
	Day 1	Day 3	Day 7	Day 1	Day 3	Day 7
Upset loss	-0.542** (0.214)	-0.180 (0.130)	-0.134** (0.061)	-0.045 (0.084)	-0.121*** (0.037)	-0.091 (0.135)
Close loss	-1.184*** (0.224)	0.209 (0.167)	0.163* (0.085)	0.130** (0.056)	0.109 (0.165)	0.077 (0.082)
Upset win	-0.117 (0.168)	-0.094 (0.095)	-0.029 (0.035)	0.007 (0.089)	-0.024 (0.039)	-0.018 (0.052)
N	8,765	8,765	8,765	12,127	12,127	12,127
<i>Panel C: Male vs. Female</i>						
	Male			Female		
	Day 1	Day 3	Day 7	Day 1	Day 3	Day 7
Upset loss	-0.249* (0.134)	-0.229*** (0.083)	-0.058 (0.086)	-0.299*** (0.071)	-0.039 (0.136)	-0.213*** (0.071)
Close loss	0.257*** (0.085)	0.175 (0.150)	0.106 (0.120)	-0.234** (0.109)	0.018 (0.136)	0.105 (0.092)
Upset win	-0.137 (0.106)	-0.123** (0.053)	-0.054 (0.073)	0.033 (0.081)	0.016 (0.051)	0.021 (0.035)
N	9,703	9,703	9,703	11,189	11,189	11,189

Notes: Each panel reports coefficients from separate regressions of support for non-interference with public criticism on the indicated subsample. High (low) internet use is defined as above (below) the weighted sample median. Urban (rural) is defined by hukou status. All specifications include province $\times$ year fixed effects, individual controls, media controls, and predicted-category controls. Standard errors in parentheses, clustered at the province level. Weighted by CGSS sampling weights. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

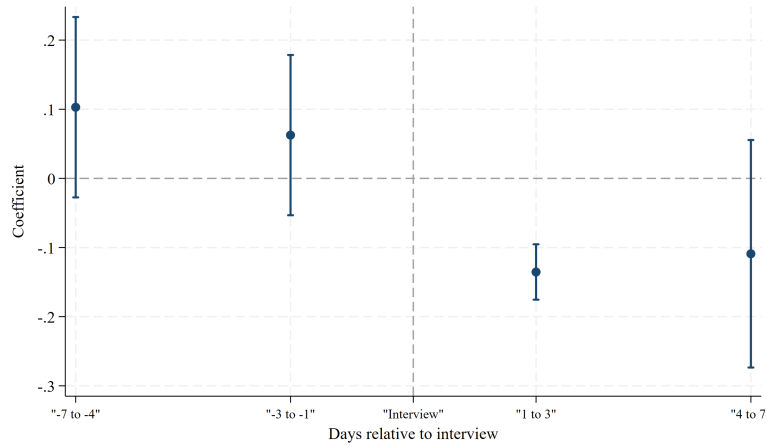


Figure 3: Event-study evidence: binned leads and lags of upset losses

Notes: The figure reports coefficients from regressions of support for government non-interference with public criticism on binned indicators for exposure to province-level upset losses relative to the interview date. Positive event times (right of the dashed line) measure upset losses occurring before the interview, in the 1-to-3-day and 4-to-7-day windows. Negative event times (left of the dashed line) are placebo lead bins and measure upset losses occurring after the interview, in the 1-to-3-day and 4-to-7-day windows. All specifications include province $\times$ year fixed effects, individual controls, media controls, and predicted-category controls, and are weighted by CGSS sampling weights. Bars show 95% confidence intervals with standard errors clustered at the province level.

5.1 Diagnostic Outcomes: Social Trust and Happiness

If the main effect reflected a broad negative-mood spillover ([Schwarz and Clore, 1983](#)), we would expect parallel declines in other attitudinal measures. Table 4 shows that upset losses have no significant effect on generalized social trust or life happiness in any window. This null result supports the interpretation that the effect is a specific shift in attitudes toward government authority rather than a general deterioration in well-being, consistent with the appraisal-tendency framework in which discrete emotions have domain-specific rather than diffuse effects ([Lerner and Keltner, 2001](#); [Meier, 2022](#)).

Table 4: Diagnostic Outcomes: Social Trust and Happiness

	Social trust			Happiness		
	Day 1	Day 3	Day 7	Day 1	Day 3	Day 7
Upset loss	0.127 (0.129)	-0.031 (0.063)	-0.018 (0.036)	0.091 (0.130)	-0.020 (0.070)	0.009 (0.053)
Close loss	-0.041 (0.094)	-0.158 (0.149)	-0.065 (0.123)	0.001 (0.053)	0.154*** (0.051)	0.080 (0.059)
Upset win	0.010 (0.049)	-0.011 (0.041)	0.003 (0.027)	0.082** (0.041)	0.012 (0.021)	0.038** (0.017)
Mean of dep. var.	3.503	3.503	3.503	3.920	3.920	3.920
Province $\times$ year FE	Yes	Yes	Yes	Yes	Yes	Yes
Individual controls	Yes	Yes	Yes	Yes	Yes	Yes
Media controls	Yes	Yes	Yes	Yes	Yes	Yes
N	17,046	17,046	17,046	19,980	19,980	19,980
R ²	0.037	0.037	0.037	0.106	0.106	0.106

Notes: Columns 1–3 use generalized social trust as the dependent variable; columns 4–6 use life happiness. Each column uses a pre-interview window $[d - W, d - 1]$, where $W \in \{1, 3, 7\}$ and d is the interview date. All columns include province $\times$ year fixed effects, individual controls, media controls, and predicted-category controls. Standard errors are clustered at the province level. Regressions are weighted using CGSS sampling weights. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.2 Alternative Fixed Effects and Province Controls

Table 5 replaces province-by-year fixed effects with separate province and year fixed effects plus province-level controls (log GDP per capita, log population, tertiary-sector share). This less saturated specification checks whether results depend on absorbing province-year variation nonparametrically. The upset-loss coefficient remains negative and significant in the 1-day window (-0.247 , $p < 0.01$) and attenuates by three and seven days, consistent with the baseline pattern.

Table 5: Alternative Fixed Effects and Province Controls

	(1) Day 1	(2) Day 3	(3) Day 7
Upset loss	-0.247*** (0.067)	-0.127 (0.081)	-0.106 (0.074)
Close loss	-0.222* (0.113)	0.012 (0.224)	-0.059 (0.142)
Upset win	-0.058 (0.074)	-0.067 (0.039)	-0.018 (0.032)
Mean of dep. var.	2.583	2.583	2.583
Province + year FE	Yes	Yes	Yes
Province controls	Yes	Yes	Yes
Individual controls	Yes	Yes	Yes
Media controls	Yes	Yes	Yes
N	20,481	20,481	20,481
R ²	0.044	0.045	0.045

Notes: The dependent variable is support for government non-interference with public criticism. Each column uses a pre-interview window $[d - W, d - 1]$, where $W \in \{1, 3, 7\}$ and d is the interview date. Columns include province and year fixed effects, province-level controls, individual controls, media controls, and predicted-category controls. Standard errors are clustered at the province level. Regressions are weighted using CGSS sampling weights. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.3 Upset-Loss Severity

Table 6 adds the cumulative goal deficit in upset losses to distinguish between the extensive margin (exposure to any upset loss) and the intensive margin (severity of the defeat). The coefficient on upset-loss count remains negative and significant across all windows (-0.435 , -0.430 , and -0.241), while the goal-deficit coefficient is not consistently negative. The result is thus driven by the occurrence of an upset loss rather than by the margin of defeat.

Table 6: Upset-Loss Severity

	(1)	(2)	(3)
	Day 1	Day 3	Day 7
Upset loss (count)	-0.435** (0.150)	-0.430*** (0.075)	-0.241* (0.133)
Upset-loss goal deficit	0.076 (0.121)	0.123*** (0.025)	0.060 (0.049)
Close loss	0.027 (0.093)	0.110 (0.142)	0.105 (0.064)
Upset win	-0.047 (0.066)	-0.057 (0.042)	-0.024 (0.037)
Mean of dep. var.	2.581	2.581	2.581
Province $\times$ year FE	Yes	Yes	Yes
Individual controls	Yes	Yes	Yes
Media controls	Yes	Yes	Yes
N	20,892	20,892	20,892
R ²	0.053	0.054	0.054

Notes: The dependent variable is support for government non-interference with public criticism. Each column uses a pre-interview window $[d - W, d - 1]$, where $W \in \{1, 3, 7\}$ and d is the interview date. In addition to upset-loss counts, the specification includes the cumulative goal deficit in upset losses within the same window. Close loss and upset win are measured as counts. All columns include province $\times$ year fixed effects, individual controls, media controls, and predicted-category controls. Standard errors are clustered at the province level. Regressions are weighted using CGSS sampling weights. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.4 Randomization Inference

We assess statistical significance using randomization inference (Young, 2019), permuting the full vector of match outcome variables across province $\times$ date cells 5,000 times while holding covariates and fixed effects fixed. The p -value is the share of permutations producing a $|t|$ -statistic at least as large as the observed one.

Figure 4 plots the permutation distributions. The upset-loss coefficient falls in the left tail of its distribution in every window ($p = 0.091, 0.076$, and 0.092), while close-loss and upset-win coefficients sit near the center ($p = 0.885$ and 0.573 in the 1-day window). That only upset losses produce tail results across all windows, while the two placebo-like comparisons do not, is difficult to attribute to chance alone.

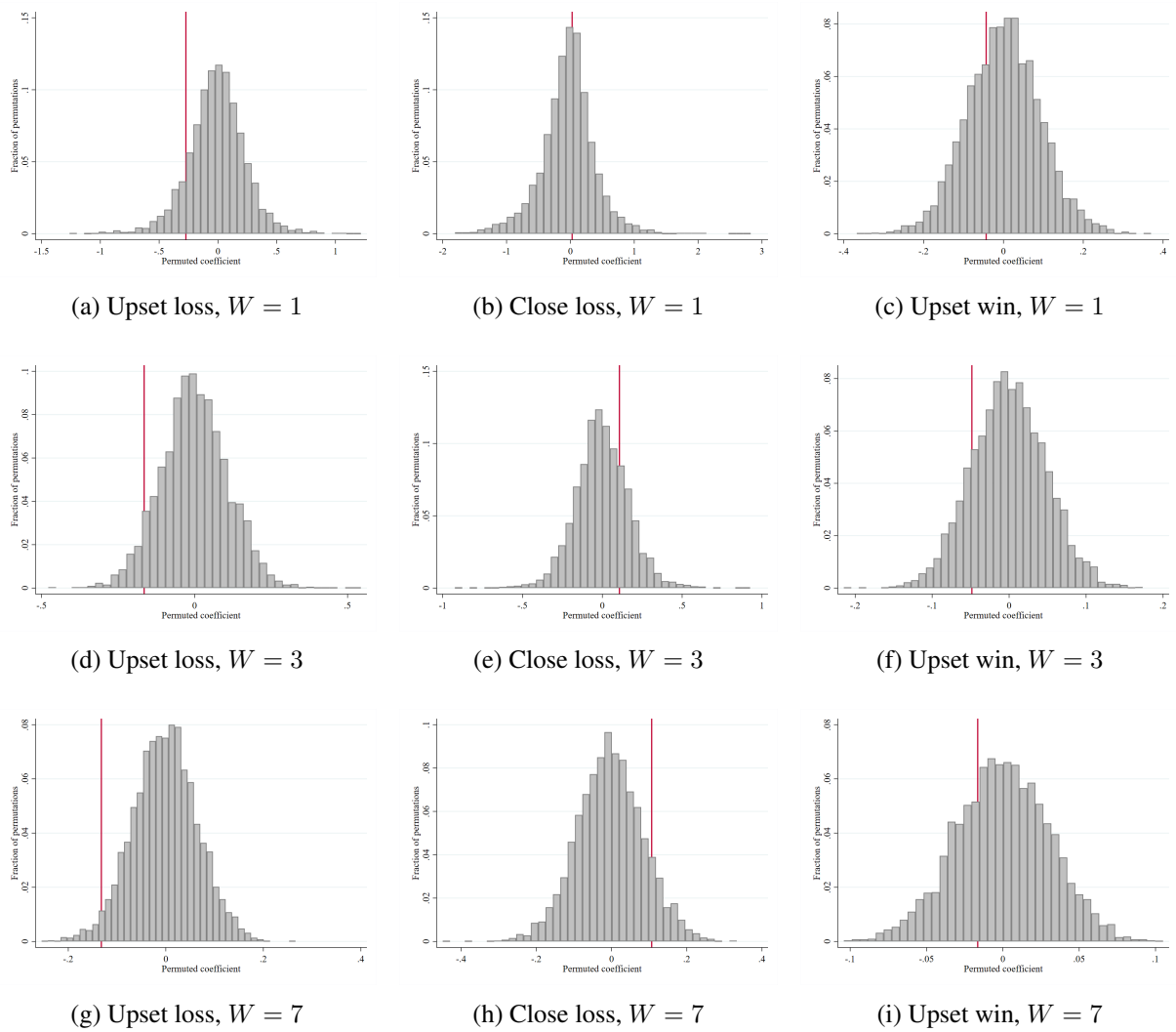


Figure 4: Randomization inference: permutation distributions of coefficients

Notes: Each panel plots the distribution of coefficients from 5,000 permutations of match outcome variables across province $\times$ date cells. The red vertical line marks the actual estimate. Rows correspond to 1, 3, and 7 day windows; columns to upset losses, close losses, and upset wins. All specifications include province $\times$ year fixed effects, individual controls, media controls, and predicted category controls, weighted by CGSS sampling weights. The p -value is the fraction of permutations yielding $|t| \geq$ observed $|t|$ (Young, 2019).

6 Conclusion

Unexpected football losses temporarily increase public acceptance of government intervention in political expression. An upset loss reduces support for non-interference with public criticism by 0.272 points (0.24 standard deviations) in the one-day window. The effect fades over the following week and is absent for expected losses and unexpected wins.

Prior work on emotional spillovers from sports finds a same-valence pattern: negative shocks produce negative downstream outcomes such as violence, harsher sentencing, and stock market declines. Our results show that this pattern does not hold in all domains. In an authoritarian context, where the state is the default provider of social order, transient negative arousal appears to increase demand for that order rather than erode trust in it. This response is consistent with the compensatory control mechanism (Kay et al., 2008), with experimental evidence that negative emotions raise demand for government intervention (Algan et al., 2025), and with the finding that rain-induced negative affect reduces support for political change (Meier et al., 2019). The institutional environment appears to mediate the *direction* in which incidental emotions spill over into political attitudes.

Political attitudes are often treated as stable dispositions, yet our results show that they respond, at the margin, to emotional cues unrelated to government performance. Standard accounts of authoritarian governance emphasize rational evaluation of government output (Fang et al., 2025; Li and Zheng, 2025) or deep cultural values (Booth et al., 2022). Zhang et al. (2025) establish a causal link from life satisfaction to political trust in China, and Zhang and Churchill (2020) show that income inequality erodes subjective wellbeing partly through neighbourhood trust, but in both cases the channel is a cognitive evaluation of life circumstances. Our results point to a more volatile component that moves with transient emotional states, and that may help explain short-run swings in public opinion otherwise difficult to attribute to policy changes.

Several limitations apply. Our estimates are intent-to-treat effects at the province level; the response among attentive fans is likely substantially larger. The randomization inference p -values (0.076 to 0.092) are close to but do not uniformly clear the 5% threshold, reflecting the rarity of upset losses in narrow time windows. The CGSS is a repeated cross-section, so we cannot track within-individual changes or test whether repeated exposure produces cumulative effects. Future work with panel data or real-time survey designs could address these limitations.

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